

Operator Discovery Report: context_scaffold_selector_bottleneck_cluster

- Operator type: scaffold_selector.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Selects the best baseline TSP scaffold based on instance structure descriptors (two-cluster bottlenecks, clustered layouts, and nearest-neighbor trap likelihood), with a safe fallback for mixed/unrecognized cases.
- Novelty classification: descriptor_based_scaffold_selector.
- Rediscovery signature: scaffold_selector:random_restart_2opt:clustered_local_search:cheapest_insertion_2opt.
- Surviving candidate: False.
- Problem class where it helps: heldout_tsplib, grid_like_tsp.

Pseudocode

- Read instance descriptors to classify the layout family.
- Dispatch the instance to the preferred scaffold for that structure class.
- Fall back to a safe default scaffold when the structure is ambiguous.

Complexity

- Non-empty lines: 13.
- Complexity score: 0.46.
- Fingerprint: 0e540a9ff56601e2.

Why It Should Help

- The operator selects among simple solvers from instance descriptors, so any gain should come from a reusable structure-to-scaffold mapping.

Failure Cases

- Underperforms on uniform_euclidean with gap delta 0.013605.
- Does not transplant cleanly into cheapest_insertion_2opt.
- Runtime inflation is high at 1.366747.

Transplant Results

- nearest_neighbor_2opt gap delta -0.035015, runtime delta 337.469929, distance-eval delta 255705.857143.
- cheapest_insertion_2opt gap delta 0.072307, runtime delta -3816.784171, distance-eval delta -6505352.142857.
- random_restart_2opt gap delta -0.014689, runtime delta 135.615186, distance-eval delta 241848.285714.
- sparse_three_opt gap delta -0.033729, runtime delta 325.875286, distance-eval delta 252284.571429.
- clustered_local_search gap delta -0.051939, runtime delta 186.963329, distance-eval delta 242675.285714.

Instance-Family Results

- heldout_tsplib operator gap 0.200044 vs baseline 0.235058.

- clustered_tsp operator gap 0.113212 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.082744 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.013605 vs baseline 0.0.

Pareto Results

- Gap delta: -0.013741.
- Runtime inflation: 1.366747.
- Distance-evaluation inflation: 26.433505.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.013741.
- flattened_selector mean gap 0.121317 and delta vs full operator 0.013741.
- shuffled_selector mean gap 0.098866 and delta vs full operator -0.00871.